

**INTERNATIONAL INSTITUTE OF MEDICAL SCIENCE
& TECHNOLOGY**

**IoT & Embedded systems organized by AIIMS Technology
Council (IIMSTC) and Engineers Council of India (ECI)**

**Project Title: Digital Adjustable LED Dimmer
using PWM NE555**

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ABSTRACT

The Adjustable LED Dimmer is an energy-efficient lighting control system designed to regulate the brightness of LEDs using Pulse Width Modulation (PWM). In this project, a PWM signal is generated using the NE555 timer IC configured in astable mode. The duty cycle of the PWM signal is controlled using a potentiometer and diode-based timing network, allowing smooth adjustment of LED brightness from low to high intensity.

The circuit operates with a 12 V DC power supply and includes resistors, capacitors, diodes, and LEDs to ensure stable operation. The diodes in the timing network separate the charging and discharging paths of the capacitor, which improves the control over the PWM duty cycle. By adjusting the potentiometer, the charging and discharging time of the capacitor changes, thereby modifying the duty cycle of the output waveform and controlling the brightness of the LEDs.

Unlike traditional dimming methods that reduce brightness by dissipating excess energy as heat, PWM dimming rapidly switches the LEDs between ON and OFF states. This technique significantly improves energy efficiency while maintaining consistent light output. The entire circuit design, including schematic diagram, PCB layout, routing, and 3D visualization, was developed using KiCad software.

This project demonstrates a simple, cost-effective, and reliable method for LED brightness control and highlights the practical application of PWM techniques in modern electronic lighting systems.

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CHAPTER 1

INTRODUCTION

Light intensity control is an important requirement in many modern lighting and electronic systems. Light Emitting Diodes (LEDs) are widely used for illumination because they are energy efficient, consume less power, and have a longer lifespan compared to traditional lighting sources such as incandescent and fluorescent lamps. In many applications such as home lighting, display systems, decorative lighting, and electronic devices, it is necessary to adjust the brightness of LEDs according to user requirements. However, traditional dimming methods that use variable resistors or linear control reduce brightness by dissipating excess electrical energy as heat, which results in energy loss and reduced efficiency.

To overcome these limitations, Pulse Width Modulation (PWM) is widely used for controlling LED brightness. PWM works by rapidly switching the LED ON and OFF at a high frequency. The perceived brightness of the LED depends on the duty cycle of the signal, which is the ratio of the ON time to the total period of the signal. By increasing the ON time, the LED appears brighter, and by decreasing the ON time, the brightness is reduced. This method provides efficient brightness control because the LED operates either in a fully ON or fully OFF state, minimizing power loss.

In this project, an adjustable LED dimmer circuit is designed using the NE555 timer IC. The NE555 timer is a versatile integrated circuit that can generate accurate timing pulses and oscillations. When configured in astable mode, it produces a continuous square wave signal that can be used to generate PWM. The circuit uses resistors, capacitors, a potentiometer, and diodes to control the charging and discharging of the timing capacitor, which determines the duty cycle of the PWM signal.

By adjusting the potentiometer, the duty cycle of the PWM signal changes, allowing smooth control of LED brightness. The circuit operates using a 12 V DC power supply, and the generated PWM signal drives the LEDs through current-limiting resistors. This project demonstrates a simple, cost-effective, and efficient method for controlling LED brightness using PWM techniques.

CHAPTER 2

PROBLEM STATEMENT

In many lighting applications, controlling the brightness of LEDs efficiently is an important requirement. Conventional dimming methods that use variable resistors or linear regulators reduce brightness by dissipating excess electrical energy as heat. This method leads to significant power loss, reduced system efficiency, and unnecessary energy consumption. As the demand for energy-efficient lighting systems increases, there is a need for a better method to control LED intensity without wasting power.

Additionally, many existing lighting control systems are complex and expensive, making them unsuitable for simple or low-cost electronic applications. Therefore, a simple, reliable, and cost-effective solution is required to adjust LED brightness smoothly while maintaining high efficiency and minimal power loss.

To address this issue, the project proposes the design of an adjustable LED dimmer using the NE555 timer IC that utilizes Pulse Width Modulation (PWM). PWM controls LED brightness by varying the duty cycle of a high-frequency switching signal, allowing efficient control of light intensity without significant power dissipation. This approach provides an energy-efficient, stable, and easily adjustable method for LED brightness control suitable for various lighting and electronic applications.

CHAPTER 3

OBJECTIVES

- To design an adjustable LED dimmer circuit using the NE555 timer IC.
- To generate a Pulse Width Modulation (PWM) signal by operating the NE555 timer in astable mode.
- To control the brightness of LEDs by varying the duty cycle of the PWM signal using a potentiometer.
- To design and implement the circuit using KiCad, including schematic design, PCB layout, routing, and 3D visualization.
- To demonstrate an energy-efficient method of LED brightness control compared to traditional resistive dimming techniques.

CHAPTER 4

METHODOLOGY

The methodology of the Digital Adjustable LED Dimmer project involves designing and implementing a Pulse Width Modulation (PWM) based LED brightness control circuit using the NE555 timer IC. The NE555 timer is configured in **astable mode** to generate a continuous square wave signal. The duty cycle of this signal determines the brightness of the LED.

The circuit consists of a timing network made up of resistors, a potentiometer, capacitors, and diodes connected to the NE555 timer. When the circuit is powered with a **12 V DC supply**, the timing capacitor charges and discharges through the resistor and potentiometer network. This charging and discharging process controls the switching operation of the NE555 timer and generates a PWM signal at the output pin.

In the circuit design, **diodes are used along with the potentiometer** to control the charging and discharging paths of the capacitor. This allows better adjustment of the duty cycle and provides smoother brightness control of the LED. By rotating the potentiometer, the ON time and OFF time of the PWM signal change, which directly controls the intensity of the LED.

The PWM output from the NE555 timer drives the LED through a current-limiting resistor. When the duty cycle increases, the LED remains ON for a longer duration and appears brighter. When the duty cycle decreases, the LED stays ON for a shorter time and appears dimmer.

The complete circuit is designed using **KiCad software**, where the schematic diagram is created and all components are connected according to the circuit design. After verifying the schematic, footprints are assigned and the PCB layout is developed. The components are arranged on the board and copper tracks are routed to form the final circuit. Finally, the design is verified using the **3D visualization feature in KiCad**, ensuring proper component placement and PCB structure before fabrication.

CHAPTER 5

COMPONENTS USED

5.1 NE555

The NE555 timer IC is one of the most widely used integrated circuits in electronics for generating accurate timing pulses and oscillations. It was originally developed in the early 1970s and has remained popular because of its simplicity, reliability, and wide range of applications. The IC contains internal components such as two comparators, a flip-flop, a discharge transistor, and a voltage divider network, which work together to produce stable timing signals.

The NE555 timer can operate in three main modes: astable mode, monostable mode, and bistable mode. In astable mode, the IC continuously generates square wave pulses without any external triggering, making it suitable for applications such as PWM generation, LED flashing circuits, and clock signals. In monostable mode, the IC produces a single output pulse when triggered. In bistable mode, it acts as a flip-flop with two stable states.

In the adjustable LED dimmer project, the NE555 timer operates in astable mode to generate a Pulse Width Modulation (PWM) signal. The duty cycle of this signal is controlled using resistors, a potentiometer, and a capacitor connected to the trigger and threshold pins of the IC. By adjusting the potentiometer, the charging and discharging time of the capacitor changes, which modifies the duty cycle of the output waveform. This PWM signal is then used to control the brightness of LEDs through a transistor driver stage. The NE555 timer is widely used in applications such as LED dimmers, motor speed control, pulse generators, frequency dividers, and industrial control systems.



Fig.5.1 NE555 timer circuit

Technical specifications

➤ Resistor R1 (10 k Ω)

Resistor R1 is connected in the timing network of the NE555 timer IC. It helps control the charging current of the timing capacitor and plays an important role in determining the frequency of the PWM signal. By limiting the current flow, this resistor ensures proper operation of the timer circuit and contributes to stable waveform generation.

➤ Resistor R2 (10 k Ω)

Resistor R2 works together with the potentiometer and capacitor to regulate the discharge path of the timing capacitor. This resistor influences the charging and discharging time of the capacitor, which ultimately affects the duty cycle and frequency of the PWM signal generated by the NE555 timer.

➤ Resistor R3 (680 Ω)

Resistor R3 acts as a current-limiting resistor for the LED. It restricts the amount of current flowing through the LED to prevent damage caused by excessive current. This ensures that the LED operates safely while producing stable light output.

➤ Potentiometer (10 k Ω Variable Resistor)

The potentiometer is a variable resistor used to adjust the brightness of the LED. By rotating the potentiometer knob, the resistance value changes, which alters the charging and discharging time of the capacitor in the timing circuit. This changes the duty cycle of the PWM signal and allows smooth adjustment of LED brightness.

➤ Capacitor C1 (0.1 μ F)

Capacitor C1 is the main timing capacitor in the circuit. It repeatedly charges and discharges through the resistors and potentiometer connected to the NE555 timer IC. This charging and discharging process determines the switching frequency and duty cycle of the PWM waveform.

➤ Capacitor C2 (10 μ F)

Capacitor C2 is used for filtering and stabilizing the power supply in the circuit. It helps remove unwanted electrical noise and ensures smooth operation of the NE555 timer IC. This improves the stability and reliability of the circuit.

➤ **Diodes (1N4148)**

The diodes are used in the timing network to control the direction of current flow during the charging and discharging of the capacitor. By separating these two paths, the diodes allow better control over the duty cycle of the PWM signal, enabling smoother brightness adjustment of the LED.

➤ **LED (Light Emitting Diode)**

The LED is the output component of the circuit and acts as the light source. The brightness of the LED depends on the duty cycle of the PWM signal generated by the NE555 timer IC. When the duty cycle increases, the LED remains ON for a longer duration and appears brighter. When the duty cycle decreases, the LED becomes dimmer.

➤ **Power Supply (12 V DC)**

The circuit operates using a regulated 12 V DC power supply. This supply provides the required operating voltage for the NE555 timer IC and other components in the circuit. A stable power supply ensures proper functioning of the PWM circuit and consistent LED brightness control.

Component	Specification	Description
NE555 Timer IC	DIP-8, 5-15 V	PWM generation in astable mode
Resistor R1	10 kΩ	Controls timing capacitor charging
Resistor R2	10 kΩ	Determines discharge timing
Resistor R3	680 Ω	LED current limiting
Potentiometer	10 kΩ Variable	Adjusts PWM duty cycle
Capacitor C1	0.1 μF	Timing capacitor
Capacitor C2	10 μF	Power supply filtering
Diodes	1N4148	Separate charge/discharge paths
LED	Standard LED	Light output device
Power Supply	12 V DC	Circuit operating voltage

Table.1 Component Specification

5.2 CIRCUIT DIAGRAM

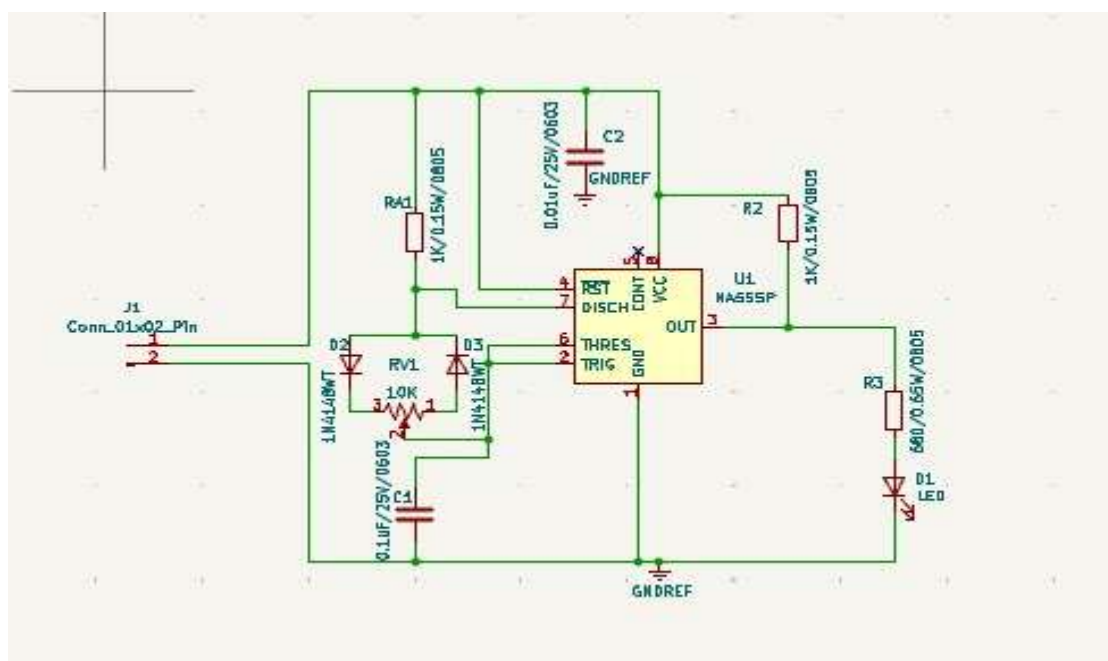


Fig.5.2.1 Schematic of Adjustable LED Dimmer in KiCad

The adjustable LED dimmer circuit is designed using the NE555 timer IC operating in astable mode to generate a Pulse Width Modulation (PWM) signal. This PWM signal controls the brightness of the LEDs by changing the duty cycle of the output waveform. The circuit mainly consists of a power supply, the NE555 timer IC, a timing network (resistors, potentiometer, and capacitor), a diode, a transistor driver stage, and LED outputs. By adjusting the potentiometer in the circuit, the charge and discharge time of the capacitor changes, which modifies the duty cycle of the PWM signal and therefore controls the LED brightness.

Main Connections in the Circuit

The NE555 timer repeatedly charges and discharges the timing capacitor through the resistors and potentiometer. When the capacitor voltage reaches the upper threshold level (about two-thirds of the supply voltage), the timer switches the output state and begins discharging the capacitor. When the voltage falls to the lower threshold level (about one-third of the supply voltage), the charging process begins again. This continuous process produces a square wave output signal.

1. Power Supply Connections

The circuit operates using a 12 V DC power supply. Pin 8 (VCC) of the NE555 timer IC is connected to the positive terminal of the power supply, and pin 1 (GND) is connected to the ground. This supply provides the required operating voltage for the timer IC and the entire LED dimmer circuit.

2. Reset and Control Pins

Pin 4 (RESET) of the NE555 timer is connected directly to the supply voltage to keep the IC continuously enabled during operation. Pin 5 (Control Voltage) is connected to ground through a small capacitor. This capacitor helps stabilize the internal reference voltage of the timer and reduces noise in the circuit, ensuring stable PWM signal generation.

3. Timing Network Connections

The trigger pin (Pin 2) and threshold pin (Pin 6) are connected together and linked to the timing network consisting of resistors, a potentiometer, and a capacitor. The capacitor repeatedly charges and discharges through this network, which determines the frequency and duty cycle of the output waveform. By adjusting the potentiometer, the charging and discharging time of the capacitor changes, which allows the duty cycle of the PWM signal to be varied and thereby controls the brightness of the LED.

4. Diode Connections

Diodes are included in the timing network to control the direction of current flow during the charging and discharging of the capacitor. These diodes separate the charging and discharging paths of the capacitor, allowing better control over the duty cycle of the PWM signal. This improves the range and smoothness of LED brightness adjustment.

5. Output Stage Connections

The PWM output signal is obtained from Pin 3 of the NE555 timer IC. This output is connected to the LEDs through current-limiting resistors. The resistors restrict the amount of current flowing through the LEDs and protect them from excessive current.

6. LED Connections

The LEDs act as the output devices of the circuit and are connected through current-limiting resistors to the output pin of the NE555 timer. The brightness of the LEDs

depends on the duty cycle of the PWM signal generated by the timer.

When the duty cycle of the PWM signal increases, the LEDs remain ON for a longer duration and appear brighter. When the duty cycle decreases, the LEDs remain ON for a shorter duration and appear dimmer.

Thus, the circuit provides an efficient method of controlling LED brightness using Pulse Width Modulation (PWM) by simply adjusting the potentiometer.

5.3 FLOWCHART

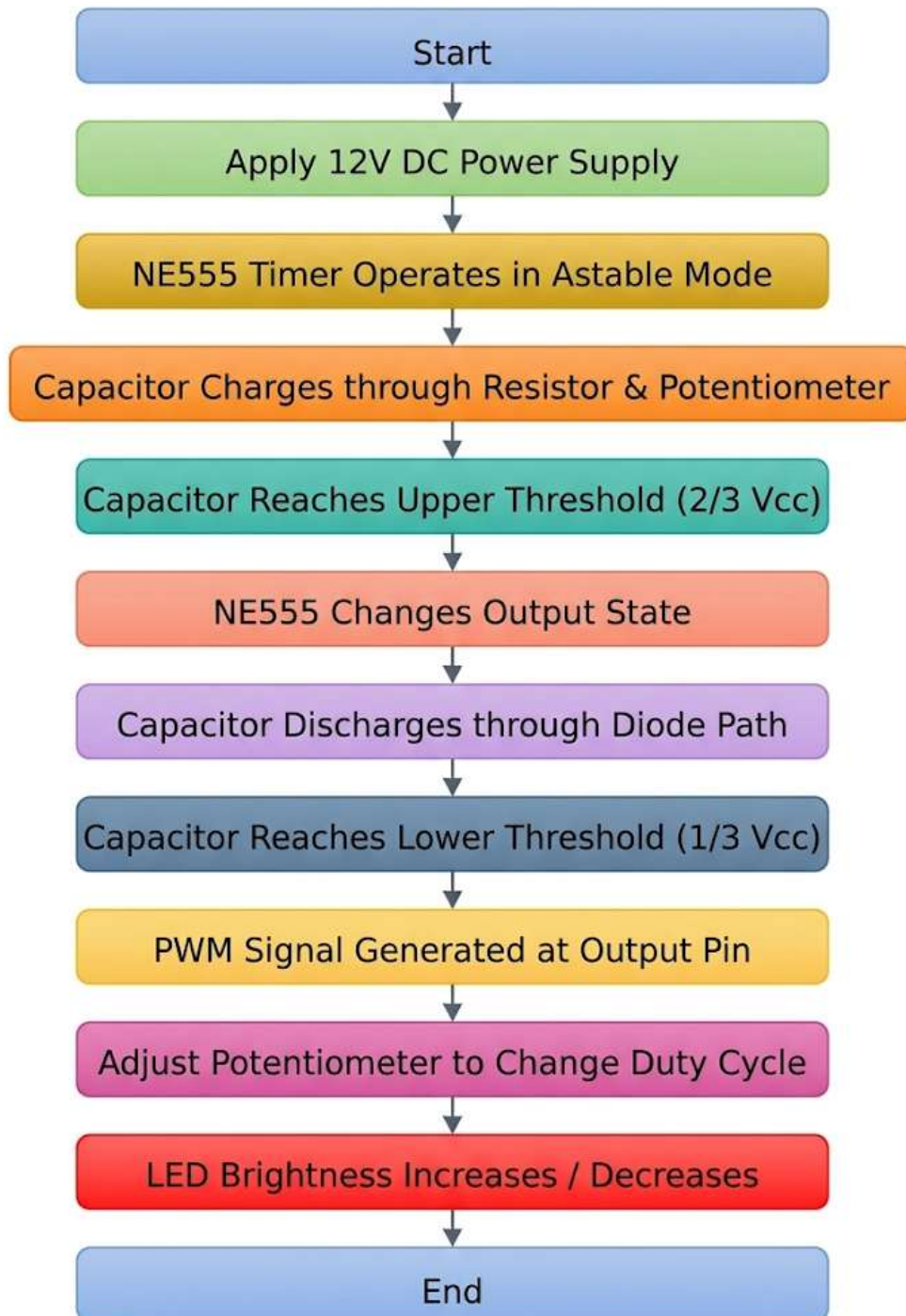


Fig.5.3.1 Flowchat

The flowchart represents the working process of the adjustable LED dimmer circuit. The system begins when a **12 V DC power supply** is applied to the circuit. The **NE555 timer IC** operates in astable mode and generates a PWM (Pulse Width Modulation) signal. The timing network consisting of **resistors, a potentiometer, and a capacitor** determines the frequency and duty cycle of this PWM signal.

CHAPTER 6

RESULTS

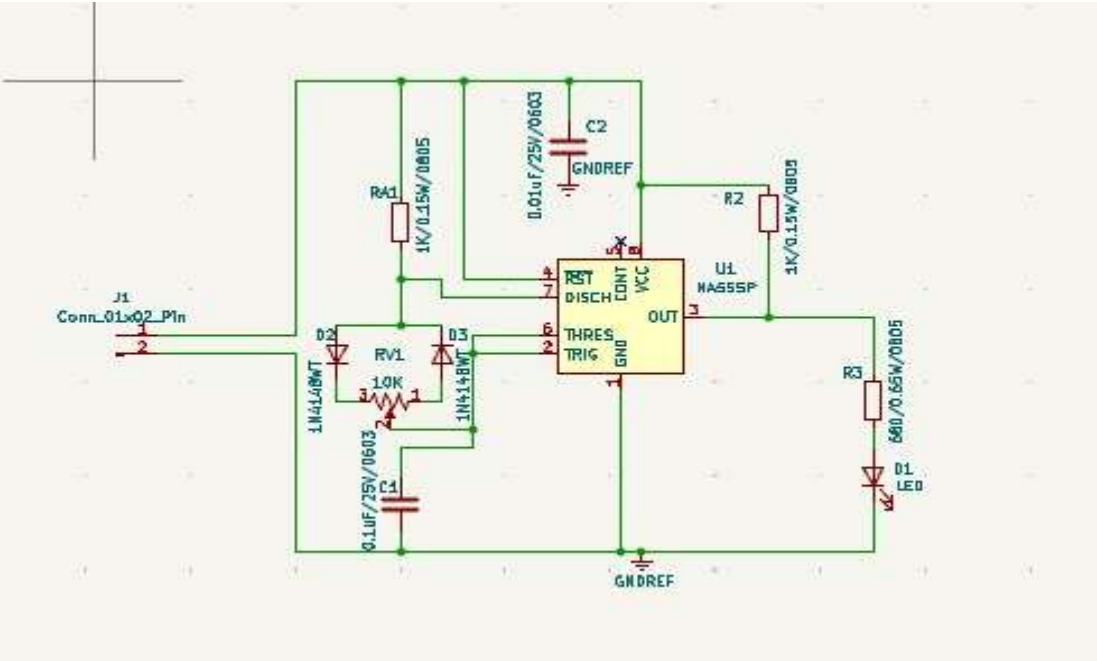


Fig.6.1 Schematic of Adjustable LED Dimmer in KiCad

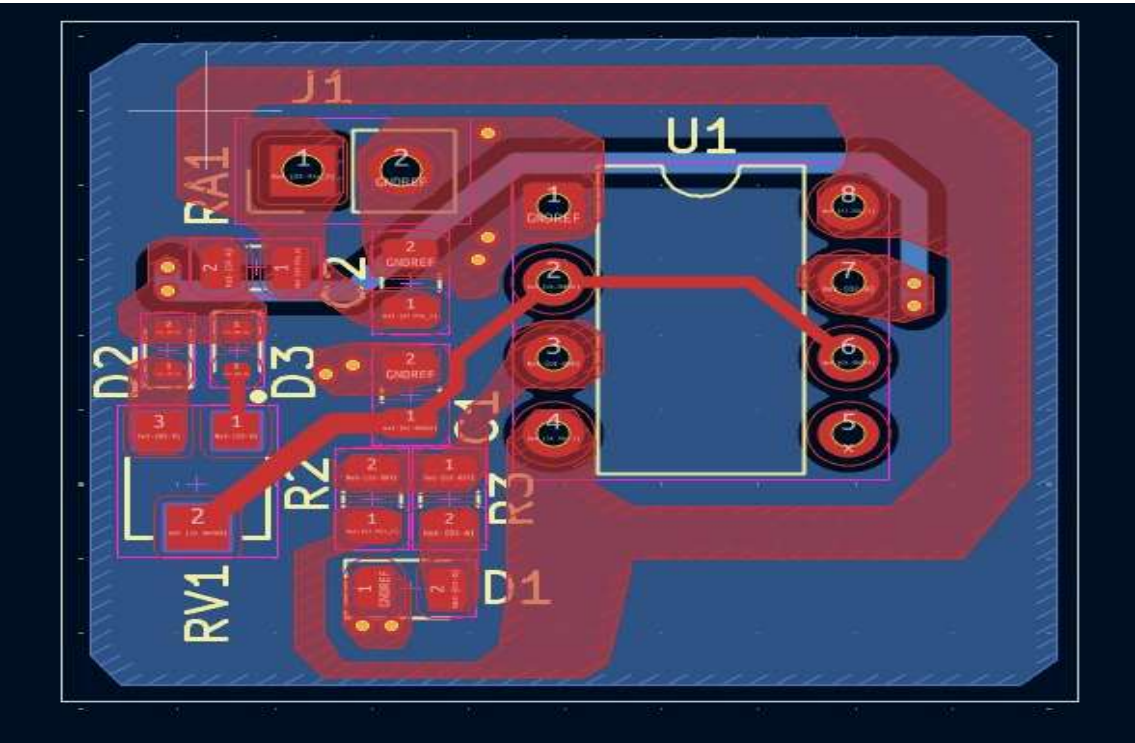


Fig.6.2 Routings of Adjustable LED Dimmer in KiCad

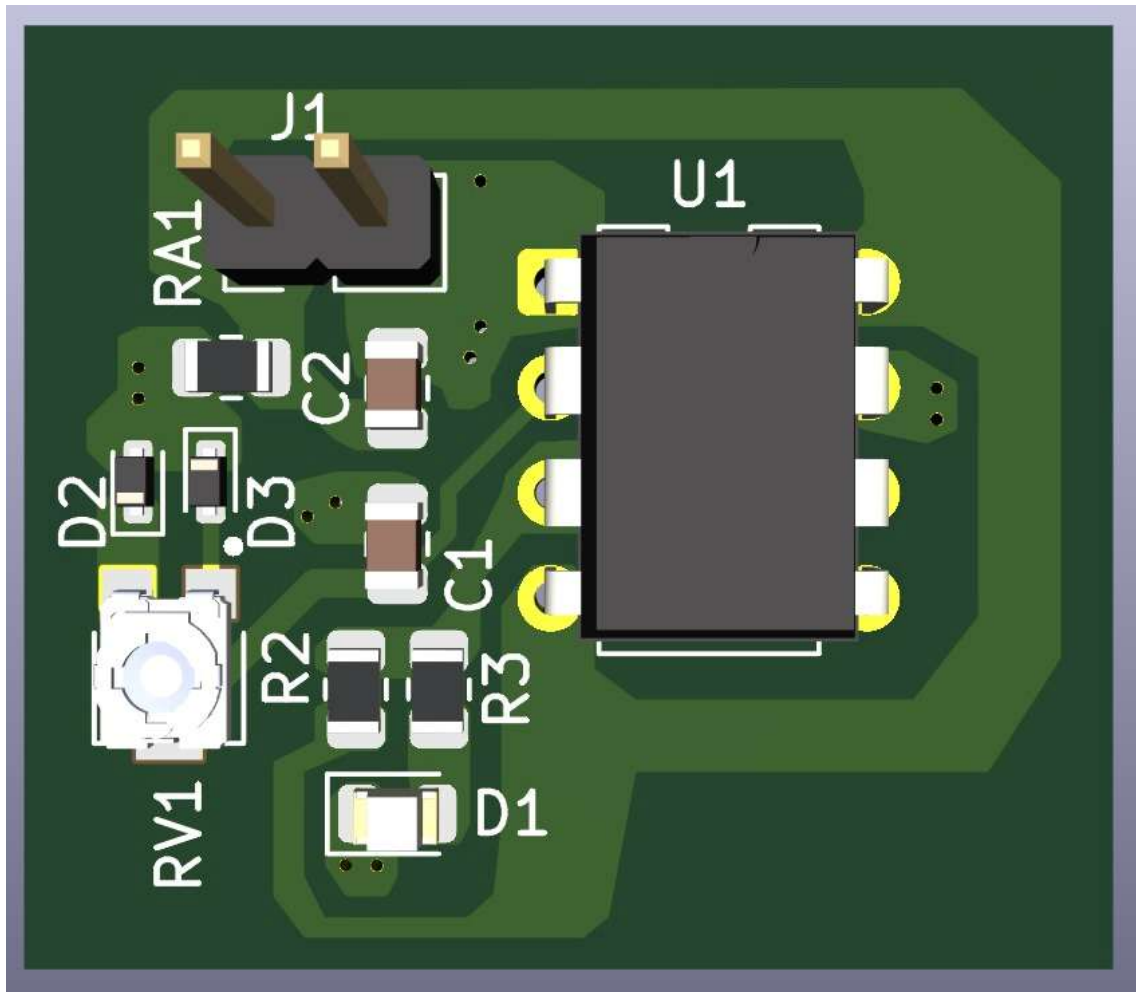


Fig.6.3 3D View of Adjustable LED Dimmer in KiCad

Schematic Design

The schematic design of the adjustable LED dimmer circuit is created using KiCad software. In this stage, all electronic components such as the NE555 timer IC, resistors, capacitors, diode, transistor, potentiometer, and LEDs are placed and connected according to the circuit diagram. Each component is assigned its appropriate symbol and electrical connections are made to represent the actual working of the circuit. The schematic helps verify that all components are properly connected before proceeding to PCB design.

PCB Routing

After completing the schematic, the next step is PCB routing. In this stage, footprints are assigned to each component and they are arranged on the PCB layout. Copper tracks are then drawn to connect the components based on the schematic connections. Proper routing ensures good electrical connectivity, reduces noise, and maintains an organized circuit layout. The routing process results in the final PCB design that can be used for fabrication.

3D View

The 3D view in KiCad provides a realistic visualization of the designed PCB. It shows how the components such as the IC, resistors, capacitors, LEDs, and transistor will appear when mounted on the board. This feature helps verify component placement, orientation, and spacing before manufacturing the PCB. The 3D view ensures that the design is practical and correctly assembled.

CHAPTER 7

SDG IMPACT ANALYSIS

SDG Impact Analysis – SDG 7 (Affordable and Clean Energy)

Sustainable Development Goal 7 focuses on ensuring access to affordable, reliable, sustainable, and modern energy for all. The adjustable LED dimmer project supports this goal by promoting energy-efficient lighting control using Pulse Width Modulation (PWM). LEDs already consume less power compared to traditional lighting systems, and using a PWM-based dimmer further improves energy efficiency by controlling brightness without wasting electrical energy as heat.

In this project, the NE555 timer generates a PWM signal that adjusts the duty cycle to control the brightness of LEDs. Instead of reducing voltage through resistive methods, the LEDs are rapidly switched ON and OFF. This method reduces energy loss and ensures efficient utilization of electrical power. As a result, the circuit helps reduce overall energy consumption in lighting systems.

Contribution to SDG 7

- **Energy Efficiency:** PWM dimming reduces power wastage and improves the efficiency of LED lighting systems.
- **Affordable Technology:** The circuit uses low-cost components like the NE555 timer, making it accessible and economical.
- **Sustainable Lighting:** LEDs combined with PWM control contribute to environmentally friendly and sustainable lighting solutions.
- **Reduced Energy Consumption:** Adjustable brightness allows users to use only the required level of light, saving electricity.

Thus, the adjustable LED dimmer supports SDG 7 by encouraging efficient energy usage and promoting sustainable lighting technologies.

CHAPTER 8

SUMMARY

The Adjustable LED Dimmer project demonstrates an efficient method of controlling LED brightness using Pulse Width Modulation (PWM) generated by the NE555 timer IC. The circuit operates in astable mode and uses a potentiometer to vary the duty cycle of the PWM signal, which allows smooth adjustment of LED brightness. A transistor is used as a switching device to control the current flowing through the LEDs, ensuring proper operation and the ability to drive multiple LEDs.

The complete circuit was designed using KiCad, including the schematic design, PCB routing, and 3D visualization of the board. The project highlights the advantages of PWM-based dimming, such as improved energy efficiency, reduced power loss, and reliable performance. Overall, the system provides a simple, low-cost, and effective solution for LED brightness control and demonstrates the practical application of electronic circuit design and PCB development.

CHAPTER 9

DISCUSSION

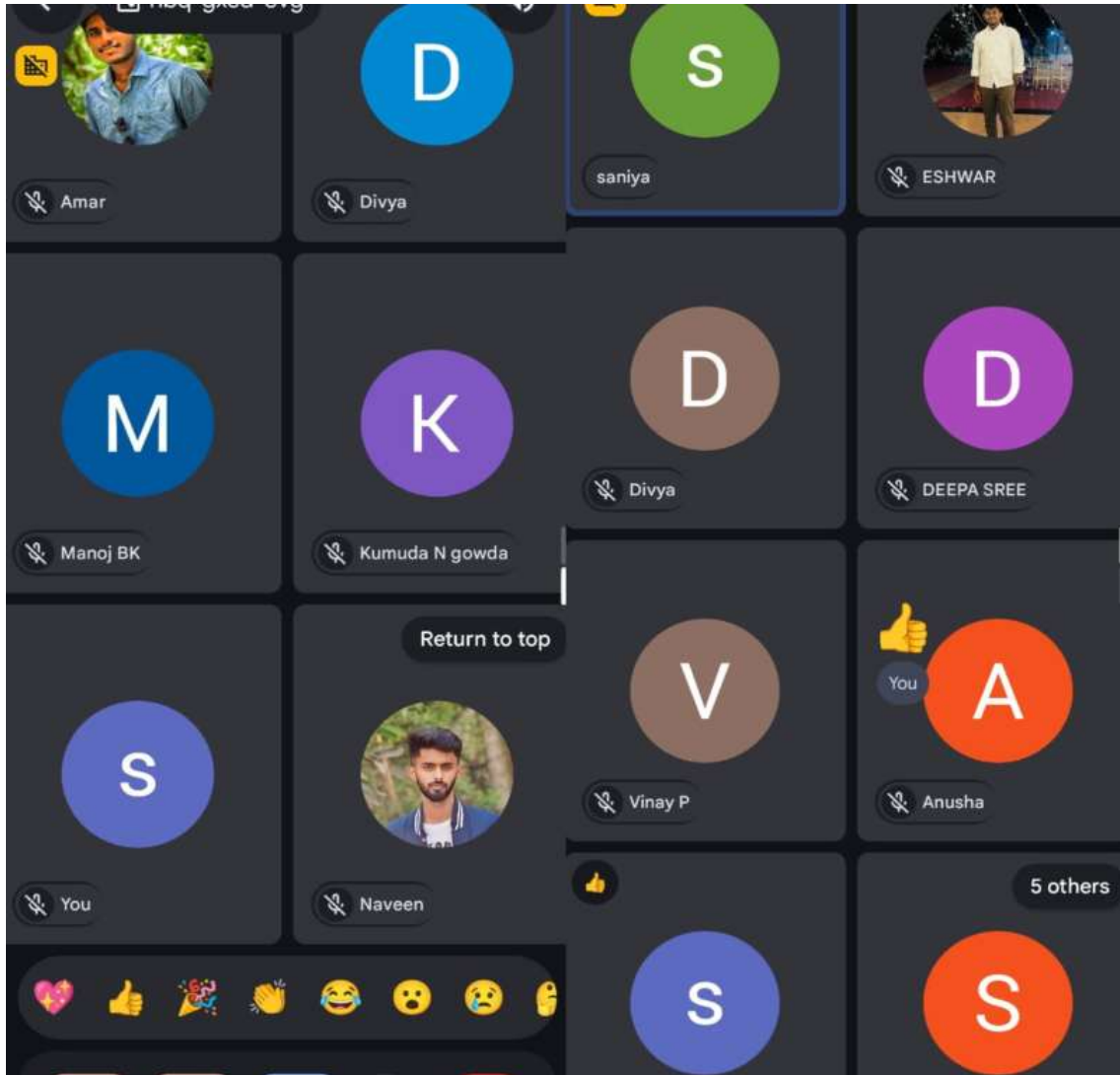


Fig.9.1 Group discussion on 6/3/2026

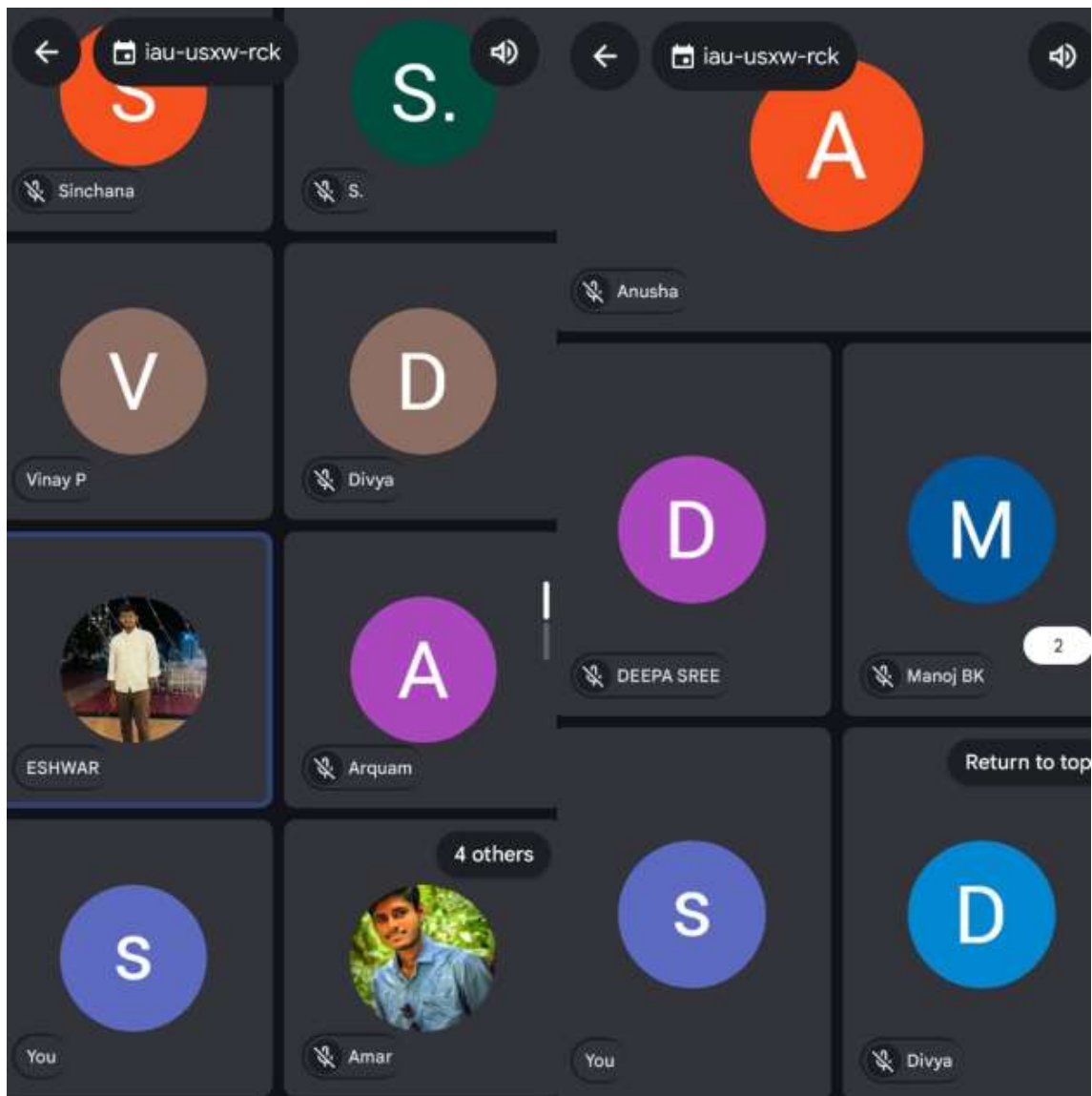


Fig.9.1 Group discussion on 7/3/2026

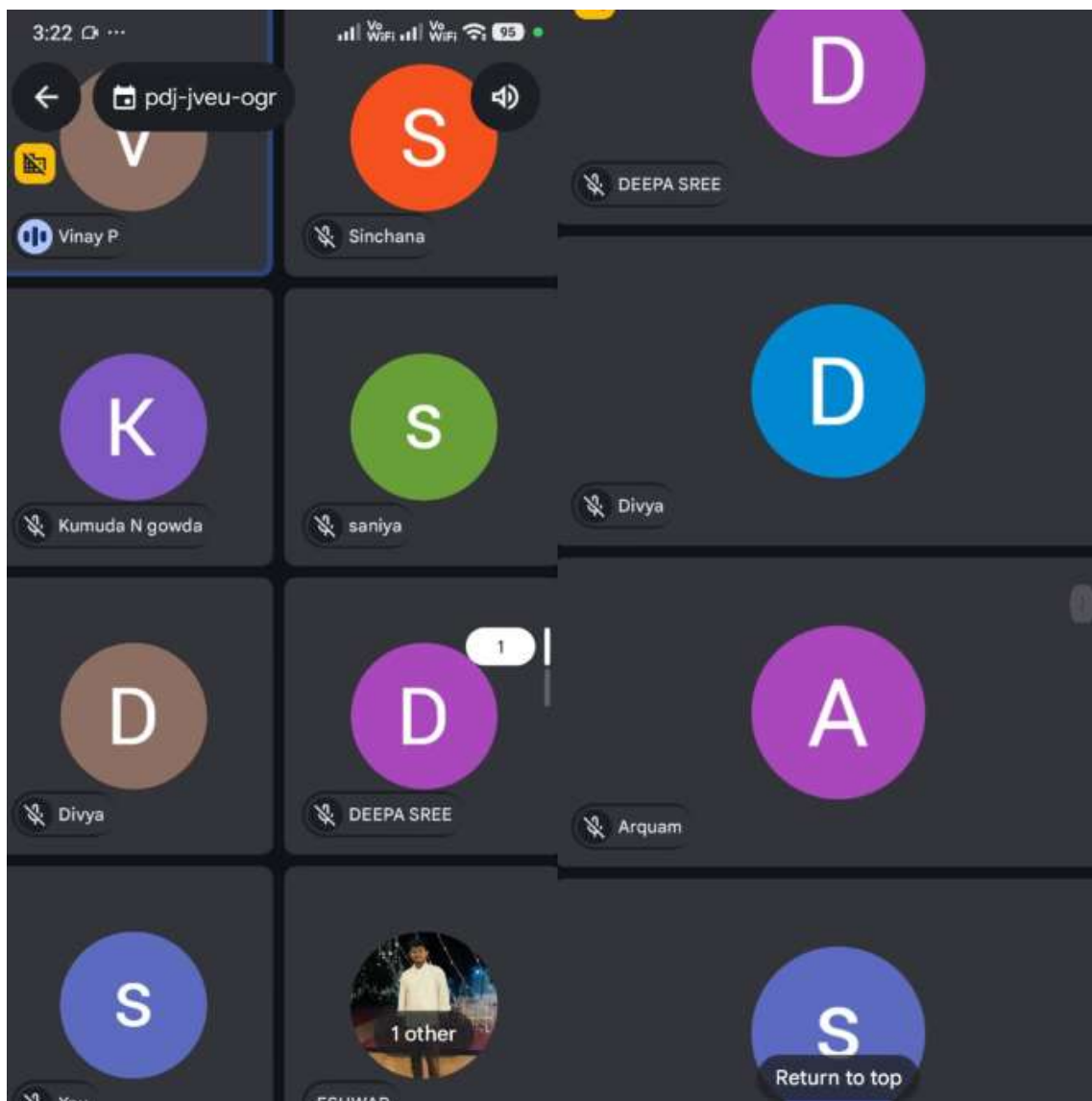


Fig.9.1 Group discussion on 9/3/2026